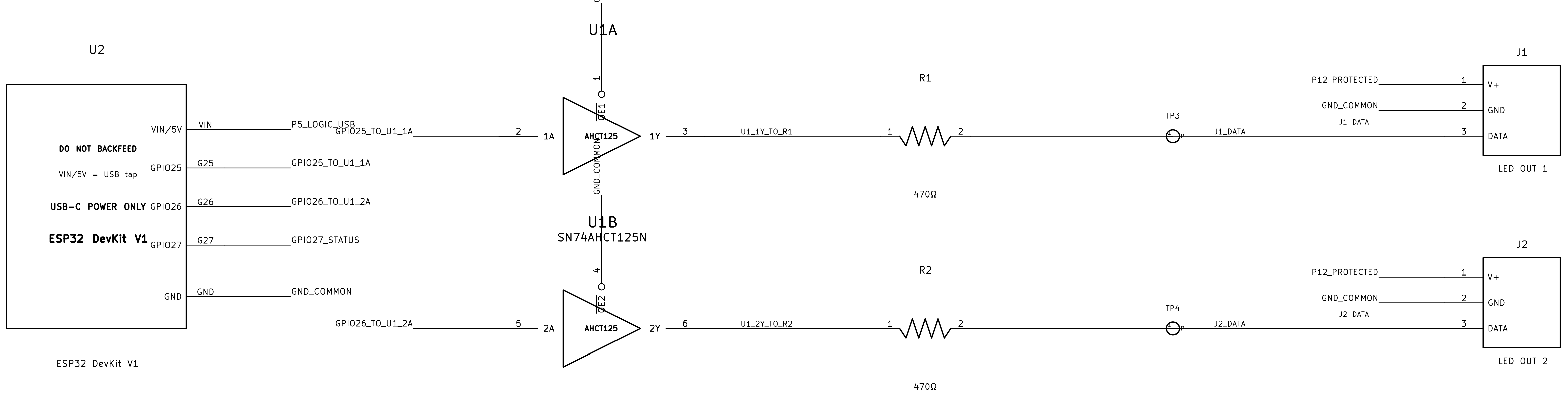
ESP32 LED Controller v6 — reviewed standard KiCad schematic

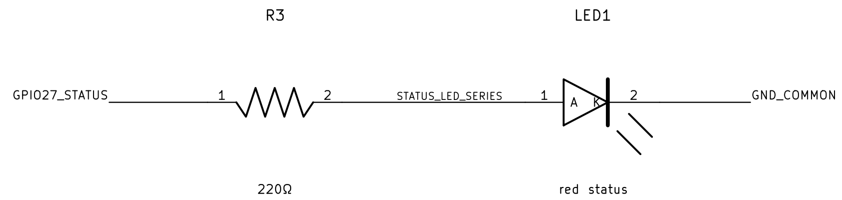
Verified netlist names match docs/esp32-breadboard/netlist.json. 12 V LED rail and USB 5 V logic rail are isolated except GND_COMMON.

CONTROL + LEVEL SHIFT — U1 SN74AHCT125N

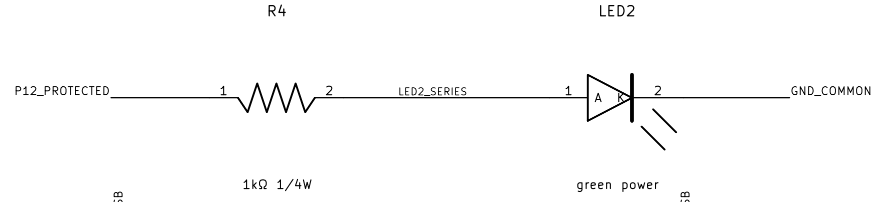
U1 = SN74AHCT125N DIP-14 quad AHCT tri-state buffer
Units: U1A/U1B used, U1C/U1D disabled, U1P power pins 14/7
OE1/OE2 -> GND; OE3/OE4 -> P5_LOGIC_USB.



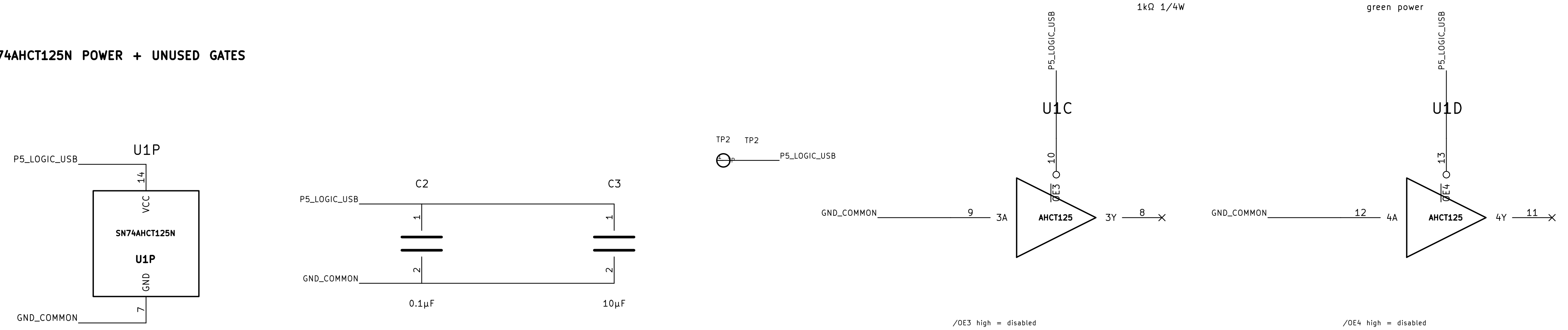
ESP32 STATUS LED



12 V POWER-GOOD LED



U1 SN74AHCT125N POWER + UNUSED GATES



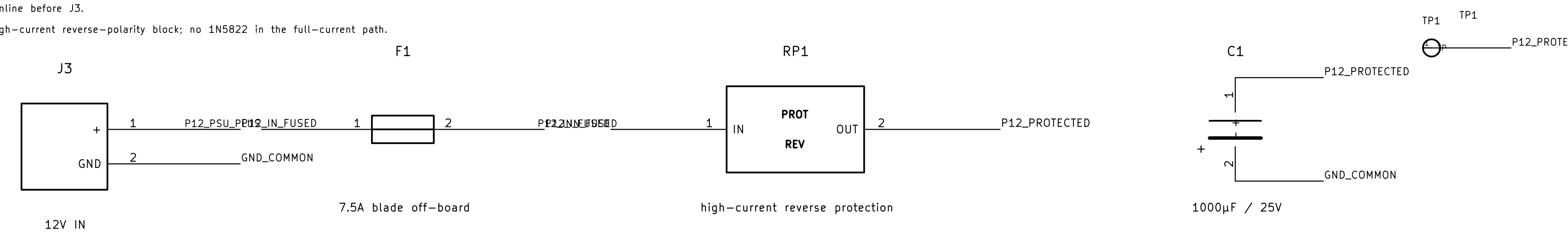
/OE3 high = disabled
A pin 9 tied low; Y pin 8 NC

/OE4 high = disabled
A pin 12 tied low; Y pin 11 NC

U1P: pin 14 VCC, pin 7 GND

12 V POWER INPUT / PROTECTION — HIGH CURRENT OFF-BREADBOARD

F1 is off-board inline before J3.
RP1 = verified high-current reverse-polarity block; no 1N5822 in the full-current path.



Pre-power: P12_PROTECTED≠P5_LOGIC_USB open/megaohms.
Common GND only; never feed ESP32 VIN from 12 V.

RP1 high-current reverse-polarity protection replaces rejected 1N5822 full-current path
U1 is SN74AHCT125N shown as U1A/U1B/U1C/U1D/U1P; net names match audited JSON netlist
Lisbon AV installation
Sheet: /
File: esp32-led-controller-v6-standard.kicad_sch
Title: ESP32 LED Controller v6 — Reviewed Standard Schematic

Size: A3	Date: 2026-05-20	**Rev: v6-kicad-standard-2**
KiCad E.D.A. 10.0.3		Id: 1/1